



Estd. 2000

ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Make-Up Test (Session: 2025-26)

Roll No.:

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Course Name: B.Tech.

Subject Name: Basics of Mechanical and Automation Engineering

Subject Code: 25ME101

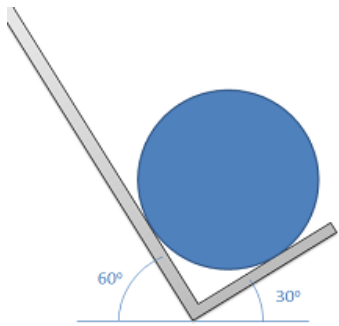
Max. Marks: 15

Time: 75 Minutes

Semester: I

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Maximum Knowledge Level: K2)				
Q. No.	Question	CO	KL	PI
1. Attempt All Questions		(1 x 6)		
(a)	What are various characteristics of a force?	CO1	K1	1.2.2
(b)	Define Volumetric stress and Modulus of Rigidity.	CO1	K1	1.2.3
(c)	State Zeroth Law of thermodynamics, and its significance.	CO2	K1	2.4.1
(d)	Define 1 Tonne of refrigeration. Write its value in kJ/min.	CO2	K2	2.4.4
(e)	What is the function of Piston rings in an IC engine?	CO3	K2	3.2.1
(f)	Write two advantages of EV cars as comparative to petrol/diesel cars.	CO3	K1	3.3.2
SECTION-B (Minimum Knowledge Level: K3)				
2. Attempt Any one Question		(Same K Levels) (3 x 1)		
(a)	State Lami's Theorem. A roller of weight 888 kN rests in an L-shaped support-frame as shown in the figure. Determine the contact reactions at all the points of contact. 	CO1	K3	1.2.2
(b)	What is Bulk Modulus? A square bar 55 mm x 55 mm is subjected to a compressive load of 600 KN. The contraction over 200 mm length is 0.45 mm and increase in thickness is 0.03 mm. Calculate the value of Poisson's ratio, Young's modulus and Bulk Modulus.	CO1	K3	1.2.3
SECTION-C (Minimum Knowledge Level: K3)				
3. Attempt Any one Question		(Same K Levels) (3 x 1)		
(a)	Enlist various main components of a domestic refrigerator and discuss its action with a neat and clean labeled diagram.	CO2	K3	2.4.3, 2.4.1

(b)	Explain 2 nd Law of Thermodynamics using various scientific statements and PMM-2.	CO2	K3	2.4.4, 2.4.2
SECTION-D (Minimum Knowledge Level: K3)				
4. Attempt Any one Question		(Same K Levels) (3 x 1)		
(a)	Explain the working of a 2-stroke Diesel engine with suitable diagrams.	CO3	K3	3.4.2, 3.4.4
(b)	Explain the working of a series and parallel types of HEVs.	CO3	K3	3.2.1, 3.3.1

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create